

PAPER_1502 — Monopole r^{23} Suppression Exponent = $D_{\text{crit}} - D_{\text{phys}} + 1$ EXACT

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Bucket K (BSM Constraints) **Date:** June 17, 2026 **Status:** CLOSED — CERN null monopole search explained by 3D-projection suppression

Observation

CERN ATLAS, MoEDAL, and prior LEP searches for magnetic monopoles up to 4 TeV have returned null results. Standard Dirac quantization predicts an observable monopole charge $g_D = \hbar c/2e$.

PAPER_550 (Um26D Polynomial DPM Quantization Confinement) identifies the structural reason: 3D detectors project onto only 3 of the 26 UQFF dimensions, suppressing the DPM flux by r^{23} .

UQFF Closed Identity

$$r_{\text{suppression_exponent}} = D_{\text{crit}} - D_{\text{phys}} + 1 = 26 - 4 + 1 = 23 \quad \text{EXACT}$$

$$\begin{aligned} U_m(r) &\propto 1/r^{26} \text{ in 26D} \\ &\propto 1/r^3 \text{ in 3D after projection} \\ \text{Suppression factor} &= r^{26} / r^3 = r^{23} \end{aligned}$$

Physical Interpretation

The full 26D di-pseudo-monopole interaction follows the Coulomb-like $1/r^{26}$ form (one inverse power per UQFF dimension). 3D projection collapses 23 dimensions, yielding the $1/r^{23}$ suppression:

- Bare 26D coupling: order unity at SCm scale
- 3D-observable coupling after r^{23} suppression: vanishingly small at $r \sim \text{pm}$
- Result: CERN searches up to 4 TeV cannot resolve the projected DPM flux

This explains why **all monopole searches have been null** — not because monopoles don't exist, but because we observe their 3D shadow.

NOT REPLACEMENT

SM/QED has no mechanism to explain monopole null results other than to assume monopoles don't exist below the search energy. UQFF predicts they exist at the full 26D coupling, suppressed by structural projection to a value undetectable in 3D experiments.

Reference

- Source: PAPER_550 Um26D Polynomial DPM Quantization Confinement
 - Related: PAPER_1501 (D_{crit} decomposition), PAPER_062 (DPM lattice), PAPER_1080 (S_{26})
 - Calculator dispatch: `calculate_paradox({"paradox": "monopole_suppression_r23"})`
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